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Temporal Evolution of the In-Situ Contact Angle in Tight Sandstone During Fracturing Fluid Exposure

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Abstract

While mature conventional reservoirs are in decline, unconventional tight sandstones and other plays are increasing in significance, but their properties such as low porosity and permeability make them economic only if stimulation techniques such as hydraulic fracturing are applied. Wettability, one of the variables controlling fluid flow and recovery in such schemes, can be altered in this process as injected fluids react chemically with the rock.

Here, we investigate the modification of the wettability in a Berea sandstone core that was initially oil wet, as a reservoir analogue to tight sandstones, all of this by sequential injecting with brine, crude oil, and broken hydraulic fracturing fluid (bHFF). The same sample was imaged several times at four critical points by high-resolution X-ray micro-computed tomography (μ CT), exposing pore-scale fluid configurations, contact angle distribution, and oil cluster connectivity in conditions that are representative of a shut-in process.

Measurements indicate that while the mean contact angle increased from $\sim 42^\circ$ for the native oil-wet state to $\sim 88^\circ$ after exposure to bHFF, consistent with the marked trend towards mixed- or water-wet behavior, there was still significant local heterogeneity. Individual oil clusters were still trapped, demonstrating that even large-scale wettability alteration is unable to eliminate residual oil entrapment, and that pore geometry and fluid flow pathways are crucial to such entrapment. Through imaging the transitions during different stages in real-time, this research demonstrates how dynamic fluid redistribution and wettability can be determined with reproducible μ CT scans providing additional insight beyond conventional wettability measurements and facilitating the design of better fracturing fluids and shut-in periods for unconventional reservoirs.

Introduction

As the world's energy consumption keeps growing and most of the conventional hydrocarbon reservoirs have been being depleted, the energy industry has increasingly shifted its focus toward the development of unconventional reservoirs (Igwe, 2024; H. Li et al., 2023). Among these, tight sandstones are quite common

and are typically characterized by porosities below 10%, permeabilities less than 0.1 mD, and pore throat sizes ranging from the nano- to micro-meter scale (Jiang et al., 2015; Tao, 2012). These formations also exhibit poor connectivity and complex, heterogeneous pore geometries (Gao et al., 2023; Jiang et al., 2015). These characteristics present significant challenges to fluid flow, hydrocarbon displacement, and overall oil recovery during hydrocarbon production (Jiang et al., 2015; H. Li et al., 2023; Tao, 2012). Hydraulic fracturing has to be generally performed in these rocks to physically break the rock and generate high-permeability fluid flow pathways to enable commercial production (Sheng et al., 2023).

One of the main properties that govern fluid behavior within such complex systems is wettability, which is defined as the rock's affinity for one fluid over another immiscible fluid (Liang et al., 2023; Yang et al., 2025). Wettability is a critical factor in multiphase flow, influencing oil–water distribution, capillary pressure, relative permeability, displacement efficiency, and residual hydrocarbon saturation (G. Liu et al., 2023; Z. Wang et al., 2022). Depending on the rock–fluid interactions, rocks can be either water-wet or oil-wet or mixed-wet, which is the case for most tight sandstone reservoirs (Liang et al., 2023; Wang et al., 2022).

One of the foremost method used to quantify wettability is contact angle due to its simplicity (Huhtamäki et al., 2018; Zhao & Jiang, 2018). However, this method is subject to a large range of error, generally bigger than other methods like Amott–Harvey and USBM (C. Liu et al., 2025).

By contrast, X-ray micro-computed tomography (μ CT) has become a successful non-destructive wettability characterization technique for more determining the contact angle in reservoir-representative conditions (Scanziani et al., 2017). This technique provides high-resolution in-situ fluid distributions and direct pore-scale measurement, while preserving core integrity and enabling repeatable analysis across multiple experimental stages (Katsamenis et al., 2019).

However, as AlRatrou et al (2017) expressed in his work, this method can be tedious due of all the image segmentation that has to be done. The post-data analysis directly impacts the accuracy of contact angle measurements and the interpretation of wettability evolution (Scanziani et al., 2017). Different techniques and artifacts might be used to correctly segment the images and achieve more accurate results. Poor segmentation can also lead to substantial fluid-phase reconstruction errors.

Following fracturing, the injected fluid is left in contact with the rock formation during a shut-in period, allowing time for pressure rebalancing and physicochemical interactions between the rock and fluid, which can modify critical reservoir properties such as wettability (Q. Wang et al., 2022). Under reservoir conditions, the fracturing fluid is degraded rapidly because of the breakers within its composition and finally will break (broken hydraulic fracturing fluid). The period for this might go from days to months, is among the most important operational parameters in hydraulic fracturing design (X. Li et al., 2025). In accordance with many studies, greater shut-in periods have potential for enhanced fluid imbibition by capillary forces, enhanced fluid diffusion distance, and overall enhanced oil recovery (Duan et al., 2025).

The main objective of this paper is to study the evolution in the wettability in tight sandstones in the post-fracturing shut-in period. The motivation for this comes from the fact that during the shut-in period, which can last from a few days to multiple months, the hydraulic fracturing fluid is degraded and can alter the wettability of the rock resulting in water blocks being formed.

Materials and Methods

Rock and fluid preparation

Rock sample. A single Berea sandstone specimen was used for the whole experiment as an analogue sample of tight sandstones, due of its relatively homogeneous and well characterized structure (Øren & Bakke, 2003); with a length of 10 mm and a diameter of 3 mm, this small core was extracted by using a core-cutting machine for more precision and avoid any kind of damage. Once we got the sample, a thin

sandpaper was used to give smoothness and the regular cylindrical shape needed. Next, the sample was soaked in deionized water to remove any contaminant. After that, a lab oven was used to dry the sample at 90°C for 24 hours. The sample, once cleaned and dried was meticulously weighted with a high precision scale.

To facilitate controlled fluid injection, this cylinder was carefully attached in the edges with small stainless ports. Once these last were in place, the whole sample was wrapped with a heat shrink tube, followed by a coating with epoxy resin to avoid any leaking.

Brine. A synthetic brine was used in the experiment, mimicking the one that is within the reservoir. Different salts, organics and inorganics components were mixed among deionized water. After that, this mixture was appropriately filtered with help of a thin filter paper, to avoid any scaling problem when pumping into the sample.

Hydraulic fracturing fluid. The hydraulic fracturing fluid (HFF) used in the experiment was created by mixing freshwater, bactericides, surfactants, pH buffers, clay stabilizers, gelling agent, and crosslinkers. The formulation mimics the fluid used in the field to provide realistic and representative fracture conditions.

The prepared HFF was loaded into a core holder, the same that was placed into the core flooding system so it can be subjected to reservoir conditions (high pressure, high temperature) and simulate the chemical degradation over time. At these conditions, the HFF reached effectively degradation within two days, turning into broken hydraulic fracturing fluid (bHFF), the same that was used as the main aging fluid to be used in the wettability tests (Singh et al., 2016).

Equipment. The ZEISS Xradia Context micro-computed tomography microCT was used to do all the scans needed, this system provided high image quality, reliable performance, and user-friendly operation, combined with an efficient workflow and high-throughput scanning capabilities (Carl Zeiss Microscopy GmbH, 2022)

Experimental Method

The overall experimental procedure is shown in Figure 1, where the initial condition of the sample is oil wet; to reach this condition, the coated sample was firstly immersed into a jar full with a light sweet crude oil, sealed and placed in a vacuum system operating at 200 psi and 90°C for a period of three weeks (Lin et al., 2019). After that the first μ CT scan was done.

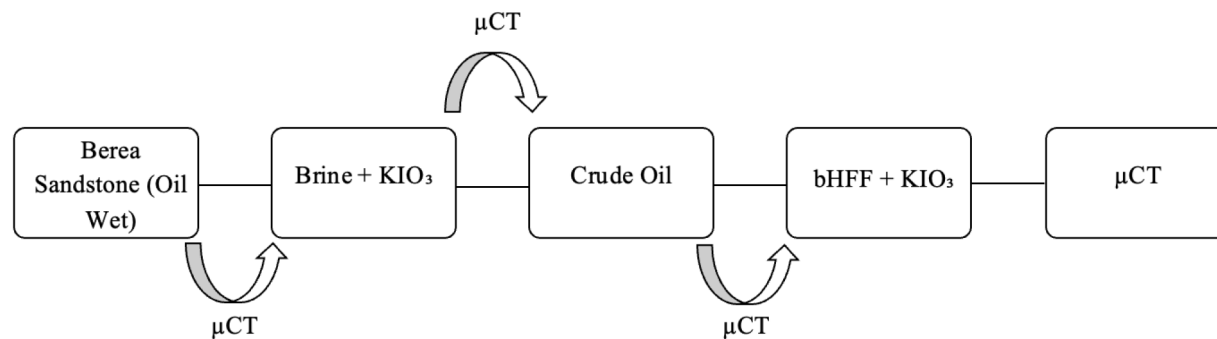


Figure 1—Experimental flow chart for the micro-scale in-situ contact angle

Contact Angle Measurement. The variation of the contact angle was measured by analyzing different μ CT scans focusing on the oil bubbles. These scans were done at the end of the following stages:

- A mixture of brine and potassium iodate (KIO_3) was pumped through the sample at a small flow rate of 50 $\mu\text{L/h}$ for a period of 24 hours, intended to simulate the spontaneous imbibition.

- The sample was removed from the scan and a second fluid was pumped, in this case, the same light sweet crude oil in which the sample was initially saturated. The conditions were the same as the first one; now a spontaneous drainage was trying to be simulated.
- In the last stage, the sample was removed again, and a third fluid was pumped, a mixture of bHFF and KIO₃, repeating the conditions as the last ones, this stage was trying to simulate the early stage of the shut-in period.

Imaging Process

An open-source image analysis package (Fiji-ImageJ) ([Schindelin et al., 2012](#)) was used to analyze the resulting data from the μ CT scans. Although care was taken to reposition the sample in the exact same spot, small alignment issues were noted which corrected by image registration. The end slices from both ends of the stack were removed as they were too bright or too dark compared with the rest. Each stack was named after their respective stage, that's it: 1.) Initial Condition, 2.) Brine Imbibition, 3.) Oil Drainage, 4.) bHFF Injection.

The 3D image stacks were converted to 8-bit grayscale form to normalize voxel intensity resolution; contrast modifications were done manually to enhance better visualization among the different phases. The scans were filtered by first adding a controlled "salt and pepper" noise, and then a median filter ($n = 4$) was used. This pre-processing step smoothed the boundaries and removed speckle noise as we can see in [Figure 2B](#).

After that, the stacks were segmented using the Trainable Weka Segmentation plugin ([Arganda-Carreras et al., 2017](#)). Due to high numbers of slices in each stack and computer memory limitation, the stacks were sectioned in sub-stacks and segmented ([Figure 2C](#)).

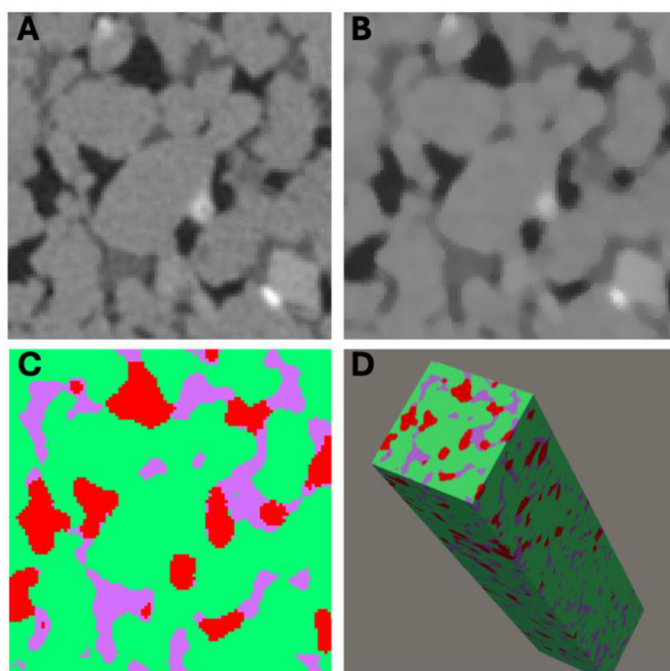


Figure 2—Image workflow of the Stage 3 (bHFF injection). A: Raw data. B: Filtered image. C: Segmented image, the green phase represents the grain and minerals of the rock; the purple phase represents the brine, and the red phase represents the oil ganglia. D: 3D representation.

[Figure 2D](#) shows the 3D visualization of one sub-stack with the three identified phases solid, oil, and bHFF represented in green, red, and purple respectively. The different contact angles were measured at the oil/solid/bHFF interfaces ([Figure 3](#)) and finally the respective comparisons and analysis were made.

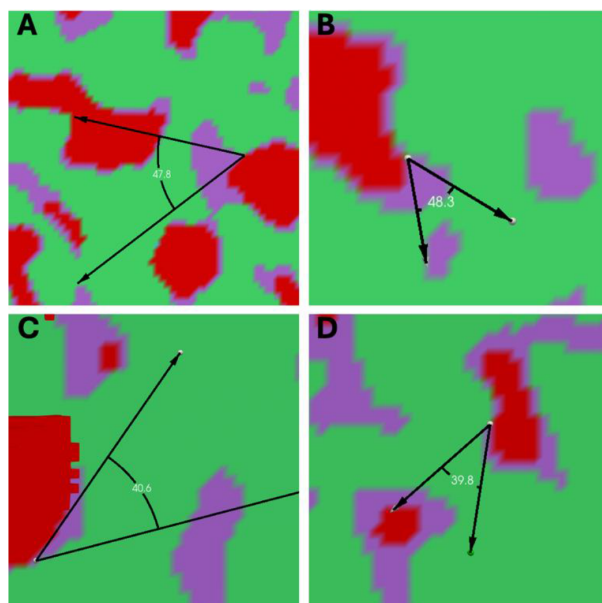


Figure 3—Contact angle measurements for Stage 3.

Results and Discussion

The Berea sandstone exhibited a porosity of approximately 12% and a mean pore size of 15 μm , with a clear presence of minerals within its structure due to the big contrast of these with the rest of the solid part, as we can see in Figure 2A. As it was expected, different parts of the sample had different wettability, usually the wetting phase tends to be in the smaller pores and the corners of the bigger pores (Andrew et al., 2014; Øren & Bakke, 2003).

As shown in Figure 4, the different contact angles were measured, and the average was calculated as well as the standard variation (error bar). The biggest oil cluster of each stage was 3D represented as well (Figure 5), to see the progression of the size and connectivity of the oil phase caused by the different exposures to fluids.

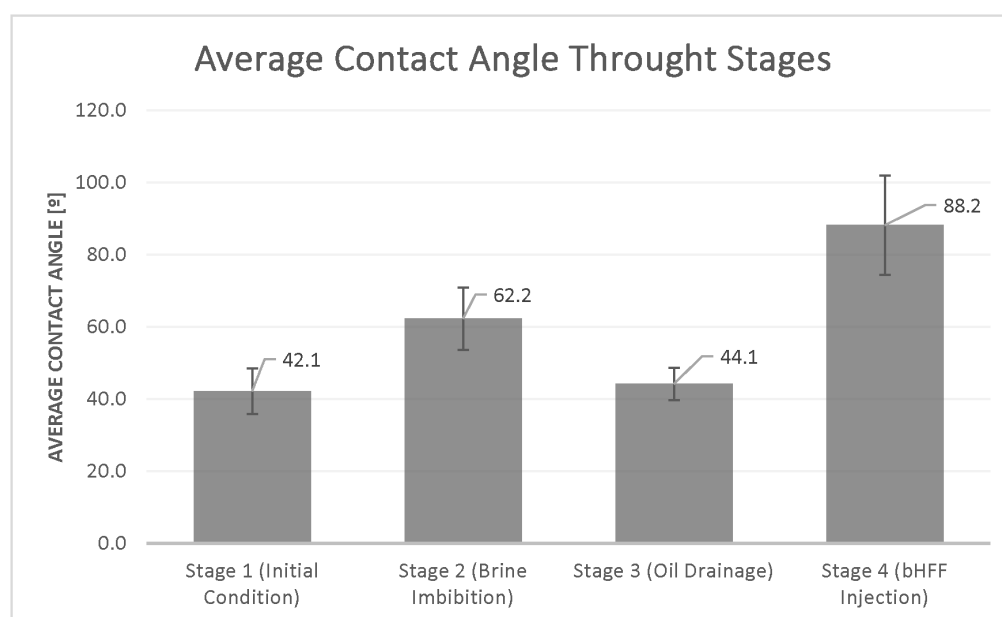


Figure 4—Average Contact Angles for the Berea Sandstone.

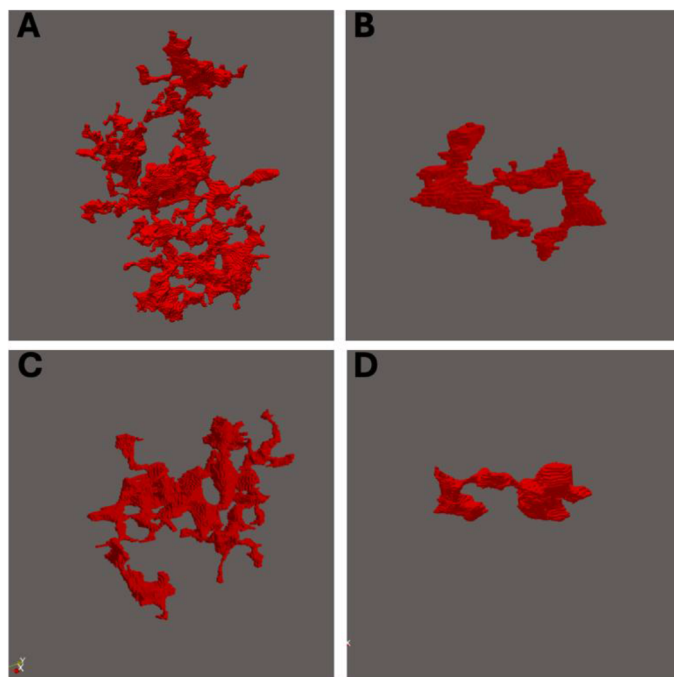


Figure 5—3D visualization of the largest connected oil ganglia identified at each experimental stage. A: Initial Condition. B: Brine Imbibition. C: Oil Drainage. D: bHFF Injection.

At the initial condition (Stage 1) the average contact angle was of $42.1^{\circ} \pm 6.3^{\circ}$; as we can see in [Figure 5](#), the main oil cluster is extremely continuous and well-connected in the pore system. These are characteristic of the strong oil-wetness of the rock, where the oil phase is an extensive coherent film that fills much of the accessible pore volume. Some irreducible water was presented in the sample as well.

In the Stage 2 (spontaneous imbibition), the biggest oil cluster is observed to decrease and to become more isolated, this indicates that spontaneous imbibition was partly successful in disrupting the film of the oil, invading few regions and creating a brine film, turning the sample into slightly intermediate wet. The average contact angle was of $62.2^{\circ} \pm 8.7^{\circ}$.

After re-injecting the original crude oil Stage 3 (spontaneous drainage), the average contact angle returned closer to the Stage 1 with $44.1^{\circ} \pm 4.5^{\circ}$, and the largest oil ganglion recovered, at least partially, its connectivity. The re-injected oil recovers part of the pore space occupied by brine, re-establishing the connection of many isolated blobs and increasing the size of the largest oil cluster.

This reconfirmed that the reintroduced oil effectively displaced the brine phase from larger connected pore throats, restoring the overall oil-wet condition. The small residual brine was persisting in some sections of the sample, however its influence on the average wettability was minimal compared to the dominant oil phase.

The greatest change was at Stage 4, where the bHFF was injected. The fluid pushed out the oil and altered the pore surface, minimizing the oil wet state of the rock. The contact angle had risen to an average of $88.2^{\circ} \pm 13.8^{\circ}$, evidently demonstrating a change away from oil-wet and towards a mixed-wet to water-wet condition. The dominant oil ganglion continues to contract and become more dispersed. This is supported by the findings of other studies where aged or broken fracturing fluids can increasingly change the surface of the rock, allowing brine to spread and the oil to retreat ([H. Wang et al., 2025](#)).

These segmented images and contact angle measurements show that changes in wettability due hydraulic fracturing process do not happen homogeneously, but instead are a mixture of pore size, exposure timing, and fluid pathway. Remaining oil clusters despite extensive fluid invasion illustrate that optimal displacement is normally limited by capillary forces, but it may be enhanced with help of some components within the fracturing fluid.

Conclusions

This study finds pore-scale wettability to be dynamic and to vary in complex ways as a reservoir analogue experiences representative fluid sequences of hydraulic fracturing and shut-in period. By observing directly how brine, oil, and degrades fracturing fluid alter fluid continuity within the same pore network, this study demonstrates that comparatively subtle chemical alterations have the potential to destroy or re-form oil pathways in ways not seen by bulk tests.

The existence of isolated oil ganglia, even in the presence of a general trend toward water-wet behavior, assure that pore geometry and residual brine can preserve mixed-wet conditions long after the average contact angle seems to change. This is why water blocks and poor flowback remain an issue in the field even if laboratory wettability experiments show a clean transition to water-wet state.

These results show that repeated scanning of a single core by μ CT at realistic exposure intervals can bridge the gap between conventional static contact angle measurement and true dynamic multiphase flow behavior. This pore-scale visualization can be applied to design the optimal chemical composition of fracturing fluids, not just for the initial steps of the hydraulic fracturing process (fractures creation), but for the interaction of the degrades fluids with rock surfaces during shut-in period and flowback.

Future studies need to follow on from this by combining pore-scale imaging with dynamic flow experiments and HFF design using shut-in periods that more closely resemble those used in field, so that the industry can better predict, and perhaps even control the change in wettability.

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